



Printing

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▼ 1. Introduction

1.1 Purpose and objectives

This report documents the methodology used to develop new Life Cycle Inventory (LCI) datasets and emission factors for textile printing processes, intended for integration into the Carbonfact database. This study adopts a transparent, literature-supported, and verifiable LCI approach.

1.2 Functional unit

The functional unit is defined as **1 kg of textile printing** (cradle-to-gate), covering core printing, drying, steaming, washing off on the specified substrate.

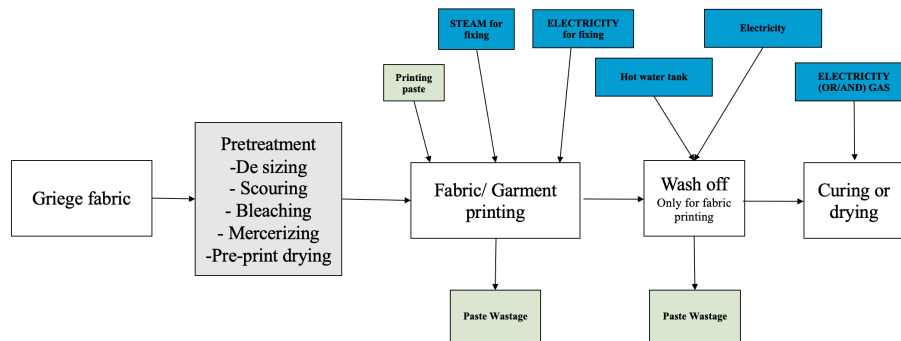
For auxiliary datasets on specialty chemicals, ink and dyes, the functional unit is **1 kg of chemical or dye, cradle-to-gate**, ready to use in textile wet processing.

For textile printing LCAs, the functional unit is often reported either as 1 kg of fabric printed or 1 m² of fabric printed. Because printing processes (flat screen, rotary screen, digital, transfer, etc.) are applied to fabrics with very different basis weights (GSM), a conversion factor between kg and m² is necessary when primary data is unavailable. The most robust methodology is to derive default conversion ratios based on fabric mass per unit area (GSM) differentiated by fibre type and typical fabric construction used in printing applications. For example, the Environmental Footprint (EF) 3.1 database commonly expresses textile processing inventories per m² of textile, while many industrial LCA datasets and mill data are reported per kg of fabric processed.

The following guidance note for [PEFCR](#) can be used for conversion if not specified as primary data.

"For textile processes that require a conversion between m² and kg, the user of the PEFCR shall use either specific grammage of the fabric or use the default value of 150g/m²" - Page 134 in PEFCR for textile and apparel.

1.3 System boundaries



Included in boundaries:

- Printing
- Drying
- Steaming
- Washing off

Excluded from boundaries (Grey) :

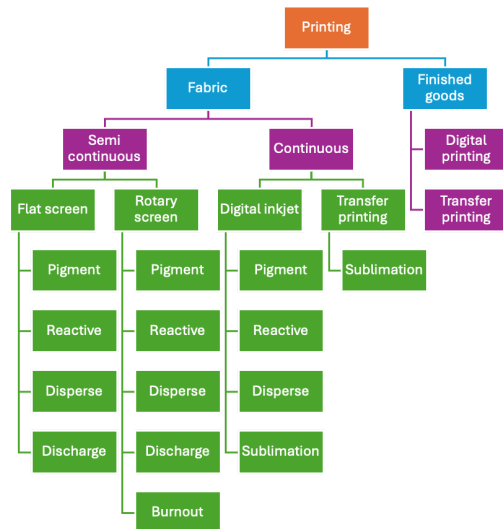
- Pre-treatment stages (upstream)
- Any post printing finishing stages other than drying and wash off

▼ 2.Classification of printing process

Textile printing can broadly be classified into fabric printing and finished goods (garment) printing, depending on whether the design is applied before or after garment manufacturing.

Fabric printing is further divided into semi-continuous and continuous processes. Semi-continuous processes include flat screen printing and Continuous processes such as rotary screen printing, digital inkjet printing, and transfer printing are designed for industrial-scale production with high throughput. Each of these methods can employ different colour systems or chemistries, including pigment, reactive, disperse, discharge, and sublimation, depending on the fibre type and fixation mechanism required.

In contrast, finished goods or garment printing occurs after the fabric has been cut and sewn into garments and is generally carried out in batch operations. The most common techniques include digital printing and transfer printing. This classification distinguishes printing technologies not only by process configuration (continuous vs batch) but also by printing chemistry in terms of dyes used in printing paste and inks.



▼ 3. Data sources and Literature review

This section highlights the data sources and the take aways from Literature on Life cycle analysis and data available for textile printing operations.

3.1 Data sources

The life cycle inventory (LCI) for textile printing was developed using a combination of secondary data sources (peer reviewed journals and supplier data sheets) to ensure both representativeness and technical robustness.

Studies such as Kujanpää & Nors (2014) and Hoque et al. (2024) offer quantitative comparisons between rotary screen and digital printing, including carbon footprint results, energy use (e.g., natural gas), and in some cases full LCI datasets covering chemicals and utilities. Peters et al. (2018) provides detailed process-level inputs for dye sublimation printing, including electricity, ink, transfer paper, and compressed air per unit area. Broader sectoral studies by Hasanbeigi (2010) and Hasanbeigi & Price (2015) contribute benchmark energy and water consumption ranges for printing and post-print washing, as well as qualitative insights into efficiency improvements. Almeida et al. (2018) focuses on energy savings potential of digital pigment printing compared to conventional methods, while Kalliala & Talvenmaa (2000) includes chemical inventory data for wet processing, including printing. Collectively, these sources enable triangulation of energy, water, and chemical inputs across technologies, supporting both attributional and comparative LCA modeling.

Chemical consumption, recipes, and dosage levels for different printing technologies (pigment, reactive, disperse, sublimation) were derived from established literature sources such as Cotton Incorporated technical manuals and United States Environmental Protection Agency reports.

Background data for upstream processes including dye manufacturing, ink production, and printing paste preparation were compiled from supplier websites, material safety data sheets (MSDS), and environmental clearance documents shared by manufacturers. This hybrid approach ensured alignment with both real-world industrial practices and published benchmarks.

3.2 EF printing dataset analysis

The EF 3.1 textile printing datasets represent multiple printing technologies such as digital inkjet (both fabric and direct-to-product), rotary/flat screen printing, transfer printing, and burnout printing standardized to a functional unit of 1 m² of printed fabric.

Across these datasets, system boundaries are relatively comprehensive, typically including pre-treatment (e.g., fabric preparation, bleaching, washing), the core printing operation, and post-treatment processes (e.g., fixing, washing, finishing, sanforizing, and quality control).

However, these datasets rely significantly on secondary sources and simplifying assumptions to generalize across technologies and geographies. For example, energy consumption values are often averaged across machine types (e.g.,

single-pass vs. multi-pass inkjet), while missing process-specific data is substituted using proxy processes (e.g., burnout printing adopting rotary screen pre/post-processing). Additionally, references are limited, and users are expected to define critical aspects such as waste treatment and transportation externally.

3.3 Takeaway from literature review

- There is a clear lack of comprehensive and transparent Life Cycle Assessment (LCA) studies focused specifically on textile printing, with very few publications providing detailed and disaggregated life cycle inventory (LCI) data for replication or benchmarking.
- Even in the available studies, transparency on inventory assumptions, chemical formulations, and process-level inputs is limited, making it difficult to validate or adapt the data for different contexts or technologies.
- Flat screen printing, despite being widely used in industry, is particularly underrepresented in literature; even basic parameters such as energy consumption, water use, and process efficiencies are rarely documented in a structured manner.
- Most existing LCA studies predominantly focus on comparisons between digital inkjet printing and rotary screen printing, creating a narrow evidence base that does not reflect the diversity of industrial printing technologies.
- Other commonly used techniques such as transfer printing, burnout printing, and discharge printing have very limited or no dedicated LCA literature, leading to significant data gaps for these processes.
- As a result, current LCA modeling of textile printing often relies on proxies, assumptions, or generalized datasets, which may not accurately capture technology-specific differences or real-world operational variability.

▼ 4. Life Cycle Inventory Modelling

This section elaborates the assumptions and data sources taken to build the LCI for textile printing datasets

Extensive documentation can be found in Annex:

- Full LCI of all printing datasets

4.1 Energy consumption

💡 Electricity consumption data were compiled from the harmonised energy database, differentiated by substrate, process technology, and dye type. The baseline electricity dataset used is `market group for electricity, low voltage (GL0)` from `ecoinvent 3.12`.

🔥 Heat consumption values were similarly compiled from the energy database. A custom heat production mix, `Heat, for dyeing process, production mix (GL0)`, was built by Carbonfact using `ecoinvent 3.12` datasets derived from the `steam production, as energy carrier, in chemical industry (RoW)` activity. The steam mix was adjusted to account for source-specific boiler efficiencies and lower heating values across the three underlying heat sources.

Key correction applied: The original ecoinvent steam production dataset included a 22% energy loss factor between boiler output and useable process steam. Since the collected LCI data represent actual fuel energy content rather than delivered steam, boiler efficiency was removed by downscaling input quantities using each fuel's efficiency factor (and LHV where necessary).

4.2 Auxiliary chemicals

Specialty chemical inventories were modelled per kg of chemical produced, cradle-to-gate, using `ecoinvent 3.12` background data and literature-based synthesis routes. Climate change impacts (kg CO₂eq/kg chemical) were computed for various chemical function categories (Thickeners, Wetting agents, Dispersing agents, Antifoaming agents, Hygroscopic agents, Reducing and oxidising agents, Binders, After washing agents).

4.3 Dye and printing ink production

Since detailed dye and ink composition and production data are not publicly available, dye and ink LCIs were developed using a bottom-up, industry-informed approach. For each dye and ink class, a representative production route was identified from environmental clearance applications and literature, and the inventory was constructed using ecoinvent proxy datasets for intermediates with comparable chemistry.

Three dye production LCIs were built:

- **Disperse dye** — azo, diazotisation–coupling route (based on Disperse Blue 183).
- **Reactive dye** — azo, monochlorotriazine route (based on Reactive Golden Yellow HER).
- **Pigment dye** — copper pthylocyanine (based on Pigment Violet 23).

Four printing ink production LCIs were built

- **Pigment ink**
- **Reactive ink**
- **Disperse ink**
- **Sublimation ink**

Pigment, burnout, discharge, and reactive printing routes have been modelled using standard industry recipes and paste formulations sourced from technical literature such as Textile Learner and Cotton Incorporated, combined with paste consumption data from the US EPA document (C4S11). The EPA source was further used to differentiate paste application efficiency between flat and rotary screen printing, ensuring realistic estimation of chemical usage across technologies.

Ink consumption for digital printing routes has been derived from a combination of patent data and BAT (Best Available Techniques) references, with all values normalized using a 150 GSM fabric assumption for consistency. Pigment ink usage is based on reported laydown ranges (0.015–0.085 g/in²), converted to 0.517 kg ink per kg fabric, with a 5% pigment concentration assumed to isolate the colorant contribution. Reactive (~18 g/m²), disperse (~14–15 g/m²), and sublimation (~10–12 g/m²) ink consumptions are taken from Textile BAT documentation, while sublimation data is further supported by the Mistra Future Fashion Report (Chalmers University). These sources collectively provide a consistent and literature-backed basis for estimating ink and chemical inputs, while acknowledging inherent variability due to fabric type, design coverage, and process conditions.

4.4 Water consumption

Water consumption for printing was compiled from process-level modelling using EPA-based paste application data and literature-derived recipes. Values show a wide variation depending on printing technology and chemistry. Reactive and disperse routes exhibit the highest water consumption (18–53 L/kg) across digital, rotary, and flat screen processes due to associated washing and fixation requirements.

In contrast, pigment, discharge, burnout, and sublimation routes show significantly lower water use (~0.04–0.53 L/kg), as they largely avoid extensive washing steps. Among technologies, flat screen processes show the highest water consumption (~52 L/kg), followed by rotary screen (~28 L/kg), while digital and transfer printing remain the lowest (<1 L/kg for most cases).

4.5 Wastewater

Wastewater was estimated assuming 10% of process water is lost through evaporation during heating and drying, with the remaining 90% discharged as effluent. At this stage, the generic **market for wastewater, average** dataset from ecoinvent 3.12 was used, as the current assessment prioritises climate change impact and variations in pollutant load have limited influence on GHG emissions. More detailed textile-specific wastewater datasets will be needed for future multi-impact assessments (eutrophication, ecotoxicity, water scarcity).

23% loss of printing paste was assumed based on US EPA document on textile printing for rotary screen and flat screen printing, however loss was considered negligible for digital printing technology. These losses are washed away and are treated as wastewater.

▼ 5. Life Cycle Impact Assessment

5.1 LCIA method and scope

Impacts were computed using Brightway2 for climate change (kg CO₂eq).

5.2 Results by printing route

Printing process route	Carbonfootprint (kgCO ₂ eq) per kg fabric
Fabric printing, Continuous, Digital inkjet, Disperse	2.39
Fabric printing, Continuous, Digital inkjet, Pigment	2.18
Fabric printing, Continuous, Digital inkjet, Reactive	2.54
Fabric printing, Continuous, Digital inkjet, Sublimation	2.54
Fabric printing, Continuous, Rotary screen, Burnout	1.65
Fabric printing, Continuous, Rotary screen, Discharge	4.84
Fabric printing, Continuous, Rotary screen, Disperse	2.93
Fabric printing, Continuous, Rotary screen, Pigment	4.21
Fabric printing, Continuous, Rotary screen, Reactive	2.28
Fabric printing, Semi-continuous, Flat screen, Discharge	7.62
Fabric printing, Semi-continuous, Flat screen, Disperse	4.02
Fabric printing, Semi-continuous, Flat screen, Pigment	6.45
Fabric printing, Semi-continuous, Flat screen, Reactive	3.28
Fabric printing, Semi-continuous, Transfer, Sublimation	2.70
Garment printing, Batch, Digital, Pigment	3.00
Garment printing, Batch, Transfer, Sublimation	3.28

Key observations:

- Digital printing (avg. 2.41 kgCO₂e/kg) is 24% lower than rotary screen (avg. 3.18) and 55% lower than flat screen (avg. 5.34), confirming the efficiency trend **Digital < Rotary < Flat** due to lower water, chemical, and energy intensity.
- **Rotary screen printing is ~40% lower than flat screen printing** (3.18 vs 5.34), reflecting better continuity, lower paste application, and higher process efficiency compared to semi-continuous flat systems.
- **Burnout printing (~1.65)** is the lowest, **~44% lower than rotary disperse (2.93)** and ~48% lower than the rotary average, because it avoids dye fixation and repeated washing, making it chemically simpler and less energy-intensive.
- **Garment printing (avg. 3.14)** is 30–40% higher than equivalent **fabric-level digital/transfer processes (2.4–2.7)**, confirming **Garment > Fabric** due to smaller batch sizes, handling losses, and lower throughput efficiency.
- **Within digital printing, pigment is ~9% lower than disperse** (2.18 vs 2.39), while **reactive is ~6% higher than disperse** (2.54 vs 2.39), due to elimination vs inclusion of washing and fixation steps.
- **Within rotary printing, pigment is ~44% higher than disperse** (4.21 vs 2.93), while **reactive is ~22% lower than disperse** (2.28 vs 2.93), indicating that curing energy dominates pigment, while optimized reactive fixation reduces impact.
- **Within flat screen printing, pigment is ~60% higher than disperse** (6.45 vs 4.02), while **reactive is ~18% lower than disperse** (3.28 vs 4.02), again showing curing vs washing trade-offs at lower process efficiency.
- **Discharge printing is ~65% higher (rotary) and ~90% higher (flat) than disperse**, because it involves pre-dyeing, chemical stripping (reducing agents), and multiple washing cycles, making it one of the most resource-intensive routes.

5.3 Contribution analysis

- Chemicals contribute 68–82% in digital printing routes (e.g., disperse :74%, pigment :75%, reactive :69%, sublimation :82%), while energy (heat + electricity) contributes only ~15–30%, showing clear chemical dominance within the defined system boundary.
- In rotary screen printing, energy dominates for disperse and reactive (50–60% combined heat + electricity), but chemicals dominate for pigment (66%) and discharge (~64%), indicating a shift depending on formulation intensity
- In flat screen printing, chemicals strongly dominate for pigment (71%) and discharge (67%), whereas disperse shows a more balanced split (~50% chemicals vs ~50% energy), and reactive still has a mixed profile with significant energy contribution
- In garment printing, chemicals still dominate but to a lesser extent (~52–55%), with energy contribution increasing (35–45%) due to lower process efficiency and higher per-unit energy use
- This overall trend higher chemical contribution in digital, pigment, and discharge routes, and relatively higher energy share in disperse/reactive rotary and flat systems is consistent with the modelling boundary where only printing, fixation, and curing are included, excluding pre-treatment and finishing stages which typically increase energy share.

5.4 Comparison with EF 3.1 datasets

Process	EF (kg CO ₂ e/m ²)	Converted using 150 GSM (kg CO ₂ e/kg of fabric)
Inkjet printing	7.11	47.40
Screen printing	13.89	92.59
Transfer printing	0.63	4.22
Digital direct-to-product	6.48	43.23
Burnout printing	7.51	50.05

- The Environmental Footprint (EF) results show significantly higher impacts (4–93 kg CO₂e/kg fabric after conversion using representative 150 GSM), indicating that the dataset represents a **full printing service system boundary**, including pre-treatment, washing, drying, and finishing, rather than isolated printing operations.
- Compared to the carbon footprint model (1.6–5.3 kg CO₂e/kg), EF values are substantially higher, reflecting the exclusion of upstream and downstream wet processing steps in the model, which are known to be major energy contributors.
- The relative trends are partially consistent, with **screen printing showing the highest impacts and transfer printing the lowest**, although absolute values differ significantly due to boundary differences.
- Transfer printing shows the closest alignment between EF and model results, reinforcing that **routes with minimal wet processing are less sensitive to system boundary expansion**, while other routes are strongly influenced by additional energy-intensive stages.

5.5 Comparison with literature sources

- Literature values indicate significantly lower impacts: Roos et al. report 2.9 kg CO₂e/kg for rotary screen (disperse), while Azizullah et al. report 1.57 kg CO₂e/kg for rotary reactive and 1.06 kg CO₂e/kg for digital reactive printing.
- The Carbonfact model results (1.6–5.3 kg CO₂e/kg) are broadly aligned in magnitude with literature, particularly for rotary disperse (~2.9) and reactive routes (~1–3), indicating consistency in process-level modelling.
- Differences across studies are primarily driven by system boundary definitions, energy mix assumptions, and level of process detail (inclusion/exclusion of washing, drying, and auxiliaries), leading to variability even within similar

printing technologies.

▼ 6. Limitations and future work

6.1 Current limitations

- Printing results are based on a **functional unit conversion from 1 m² to kg using 150 GSM**, and outcomes are sensitive to fabric weight. Printing studies often report in m², requiring careful alignment of functional units for comparison.
- Publicly available LCI data for printing-specific energy use, ink formulations, and chemical recipes are limited compared to dyeing, leading to reliance on secondary data and assumptions.
- Ink and auxiliary chemical datasets rely on generic or proxy datasets (e.g., binders, thickeners, solvents) due to proprietary formulations, introducing uncertainty in chemical contributions.
- System boundary includes only printing, fixation, and curing, excluding pre-treatment and finishing steps, which can significantly influence overall impacts in full textile processing.
- Energy consumption values for different printing technologies are often derived from literature ranges or engineering assumptions, rather than process-specific measured data.
- Limited transparency in literature on paste add-on, fixation efficiency, and process losses affects accuracy of inventory modelling across printing routes.

6.2 Future improvements

- Collect primary data from printing facilities (energy use, ink consumption, paste add-on, and machine efficiency) to improve accuracy and reduce reliance on assumptions.
- Develop technology-specific ink and chemical LCIs (pigment systems, discharge agents, binders) to better represent formulation impacts.
- Incorporate fabric variability (GSM, composition, construction) explicitly, or standardize reporting in both kg and m² to improve comparability across studies.
- Improve modelling of water and wastewater flows specific to printing processes, including chemical oxygen demand and pollutant loads.
- Introduce region-specific energy mixes and process configurations to reflect differences in industrial practices and improve geographical relevance.

▼ 7. References

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▼ 8. Annex

Full LCI of all printing datasets

Activity name	Reference product	Location	Unit	Printing classification	Process type	Process technology	Printing type
Fabric printing, Semi-continuous, Flat screen, Pigment	printed fabric, flat screen, pigment	GLO	kg	Fabric printing	Semi-continuous	Flat screen	Pigment
Fabric printing, Semi-continuous, Flat screen, Reactive	printed fabric, flat screen, reactive	GLO	kg	Fabric printing	Semi-continuous	Flat screen	Reactive
Fabric printing, Semi-continuous, Flat screen, Disperse	printed fabric, flat screen, disperse	GLO	kg	Fabric printing	Semi-continuous	Flat screen	Disperse
Fabric printing, Semi-continuous, Flat screen, Discharge	printed fabric, flat screen, discharge	GLO	kg	Fabric printing	Semi-continuous	Flat screen	Discharge
Fabric printing, Continuous,	printed fabric, rotary screen, pigment	GLO	kg	Fabric printing	Continuous	Rotary screen	Pigment

Rotary screen, Pigment							
Fabric printing, Continuous, Rotary screen, Reactive	printed fabric, rotary screen, reactive	GLO	kg	Fabric printing	Continuous	Rotary screen	Reactive
Fabric printing, Continuous, Rotary screen, Disperse	printed fabric, rotary screen, disperse	GLO	kg	Fabric printing	Continuous	Rotary screen	Disperse
Fabric printing, Continuous, Rotary screen, Burnout	printed fabric, rotary screen, burnout	GLO	kg	Fabric printing	Continuous	Rotary screen	Burnout
Fabric printing, Continuous, Rotary screen, Discharge	printed fabric, rotary screen, discharge	GLO	kg	Fabric printing	Continuous	Rotary screen	Discharge
Fabric printing, Continuous, Digital inkjet, Pigment	printed fabric, digital inkjet, pigment	GLO	kg	Fabric printing	Continuous	Digital inkjet	Pigment
Fabric printing, Continuous,	printed fabric, digital inkjet,	GLO	kg	Fabric printing	Continuous	Digital inkjet	Reactive

Digital inkjet, Reactive	reactive						
Fabric printing, Continuous, Digital inkjet, Disperse	printed fabric, digital inkjet, disperse	GLO	kg	Fabric printing	Continuous	Digital inkjet	Disperse
Fabric printing, Continuous, Digital inkjet, Sublimation	printed fabric, digital inkjet, sublimation	GLO	kg	Fabric printing	Continuous	Digital inkjet	Sublimatic
Fabric printing, Semi- continuous, Transfer, Sublimation	printed fabric, transfer, sublimation	GLO	kg	Fabric printing	Semi- continuous	Transfer	Sublimatic
Garment printing, Batch, Digital, Pigment	printed garment, digital, pigment	GLO	kg	Garment printing	Batch	Digital	Pigment
Garment printing, Batch, Transfer, Sublimation	printed garment, transfer, sublimation	GLO	kg	Garment printing	Batch	Transfer	Sublimatic